João Pedro Dudziak Fonseca

jfonseca32@gatech.edu | [Website] | [LinkedIn] | (470) 929-6290

EDUCATION

Georgia Institute of Technology, Atlanta, GA — Graduation: *May 2027*

- BS/MS Mechanical Engineering (Automation/Robotic Systems Conc.), CS Minor (Computing/Intelligence), GPA: 3.89/4.00

EXPERIENCE

Underwater Jet Propulsion Researcher at Bhamla Lab (DARPA Funded), Atlanta, GA — *May 2025 - Present*

- Conducting experimental/computational fluid dynamics research on self-designed, squid-inspired flexible nozzle to improve underwater efficiency beyond conventional propellers (η = 60% to > 80%).
- 3D Particle Tracing Velocimetry (PTV) with high-speed and laser-based visualization to track flow and nozzle deformation.
- Validating Fluid-Structure Interaction (FSI) simulation models by comparing experimental flow to numerical predictions, contributing to the development of a Physics-Informed Neural Network (PINN) optimization framework for nozzle shape.
- Using ML-based motion tracking in high-dimensional motion from live squid, informing data-driven design optimization.

Equipment Engineering Intern at Coca-Cola (2 Rotations), Atlanta, GA — *May 2023 - Aug 2023, Jan 2024 - May 2024*

- Researched and integrated palm recognition payment in vending machines, estimated to cut customer journey time by 10%. Secured a partnership for the technology, kickstarting a company worldwide project, now in market trials.
- Designed and 3D printed custom bottles with standard threading to test carbonation in dispensed liquids, automating current methods by using a vibrating table. Cut test-time in half; improved measurement accuracy by 2 decimal places.
- Developed the Technical Authorization Requirements (TAR) for 20k+ deployed Coke&Go coolers.
- Designed attachable fixture for coolers to diffuse light to minimize color degradation of cans in outdoor vending machines.
- Computed tipping angles with added forces through manual calculations and FEA for the new line of solar-powered vending machines, validating stability with factor of safety of 2 and adherence to safety standards.

Aerodynamics Engineer at GT FSAE HyTech Racing, Atlanta, GA — *Sep 2022 - May 2026*

- Designed suspension wishbones for car in CATIA; validated aerodynamic behavior with Fluent and wind tunnel testing.
- Adjusted the front-wing endplate's angle of attack by 3 degrees based on F1 aerodynamic literature, increasing downforce by 4 kg at 120 km/h with negligible drag penalty, improving stability in high-speed cornering.
- Redesigned sidepods in CATIA for a lower/slimer profile, reducing drag while maintaining efficient hot air dispersion.
- Developed and validated the chassis of the car for the first time with 3D FEA in Ansys and hand-calculations to ensure structural integrity and prevent frame failure encountered by 2022's car.
- Calculated deflection of battery box to prevent electrical shorting due to driver weight. Validated factor of safety of 1.5.

Production Engineering Intern at Sorvetes Rochinha Ice Cream CO, São Paulo, Brazil — *May 2022 - Aug 2022*

- Applied process mapping to the entire production line and identified packaging as a bottleneck.
- Reduced packaging prep time by 1 hour (per 8 hour shift) by designing a mount for an automatic labelling machine.
- Took the initiative to investigate scrap on their industrial-grade popsicle machine. Identified a wrapping material problem preventing proper heat sealing in +10% of produced units.
- Integrated ISO9000 standards into the production process to ensure adherence to quality benchmarks.

PROJECTS

Founder & Lead Engineer at TrailMate (Follower Robot Backpack), Atlanta, GA — *April 2025 - Present*

- Built an autonomous robot (DFMA+GD&T), including powertrain (geared DC motor with 1:9 transmission), chassis (modular with adjustable wheelbase/mounting systems), steering (servo-driven rack and pinion), and suspension (double-wishbone with 80 mm shock absorbers), using CAD and FEA stress, buckling and dynamics analysis in NX to support 20 kg+ payloads.
- Developed a real-time vision-based control pipeline on Raspberry Pi and STM32 by combining OpenCV background subtraction (for obstacle avoidance) and NanoDet neural network inference (for person tracking), implementing a normalized scoring algorithm to compute steering and throttle commands based on proximity and relative position.

Inverse Kinematics Approximation for Robotic Arms with Machine Learning, Atlanta, GA — *May 2025 - Aug 2025*

- Developed a machine learning framework for robotic arm inverse kinematics, achieving sub-degree joint accuracy and sub-millisecond inference for real-time, high-DOF manipulators; awarded "Best Project" among 30+ teams.
- Built supervised (MLP, SVR, Gradient Boosting) and unsupervised (K-Means, autoencoders, DBSCAN) models with PyTorch and Scikit-Learn for reachability classification and direct joint-space regression on 1.5M+ PyBullet-simulated samples; demonstrated order-of-magnitude improvements in precision and speed over analytical IK solvers.

SKILLS & HONORS

Prototyping Instructor (PI) at Georgia Tech Invention Studio - 2024

UKMT Mathematics Challenge Gold Medalist - 2020 | 2021

Engineering: SolidWorks, CATIA, Siemens NX, Blender, Ansys, COMSOL, MATLAB, Simulink, PyBullet, CFD, FEA, Thermal Analysis, Stress Analysis, Mechanical Design, Product Development, Design Validation, Root Cause Analysis, GD&T, DFMA, FMEA, BOMs, CNC Machining, 3D Printing, Microcontrollers, PLM, Heat Transfer, Dynamics, Mechanics of Materials, Vibrations

Coding: Python, Java, C/C++, JavaScript, HTML, CSS, Selenium, Anaconda, PyTorch, ScikitLearn, TypeScript, Node.js, STM32